STAT 7650. Homework 3 (Due: Tuesday, February 24, 2026)

1. (20 points) The table reports population counts of the flour beetle (*Tribolium confusum*) at various time points. Beetles at all developmental stages were included in the counts, and the food supply was carefully controlled.

Days	0	8	28	41	63	79	97	117	135	154
Beetles	2	47	192	256	768	896	1120	896	1184	1024

An elementary model for population growth is the logistic model given by

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right),$$

where N is population size, t is time, r is a growth rate parameter, and K is a parameter that represents the population carrying capacity of the environment. The solution to this differential equation is given by

$$N_t = f(t) = \frac{KN_0}{N_0 + (K - N_0)\exp\{-rt\}},$$

where N_t denotes the population size at time t. The parameter to be estimated is $\theta = (N_0, K, r)$.

(a) (5 points) Fit the logistic growth model to the flour beetle data using the Gauss-Newton method to minimize the sum of squared errors between the model predictions and the observed counts.

(b) (5 points) Fit the logistic growth model to the flour beetle data using the Newton–Raphson method to minimize the sum of squared errors between the model predictions and the observed counts.

(c) (5 points) In many population modeling applications, an assumption of log-normality is adopted. The simplest formulation assumes that $\log N_t$ are independent and normally distributed with mean $\log f(t)$ and variance σ^2. Under this assumption, find the MLEs using both the Gauss–Newton method and the Newton-Raphson method.

(d) (5 points) Under the log-normal model, find the standard errors of the MLEs and estimate the correlation between them.

2. (10 points) The observations $\{(y_i, x_i), y_i \in \mathbb{R}, x_i \in \mathbb{R}^p, i = 1, \ldots, n\}$ are sampled independently from a linear regression model,

$$y_i = \beta^T x_i + \varepsilon_i, \quad i = 1, \ldots, n.$$

The elastic net method estimates β by solving

$$\min_{\beta} \sum_{i=1}^{n} (y_i - \beta^T x_i)^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2,$$

where $\|\cdot\|_1$ and $\|\cdot\|_2$ denote the ℓ_1 and ℓ_2 norms, respectively. This formulation reduces to the ridge regression when $\lambda_1 = 0$ and to the lasso when $\lambda_2 = 0$.

(a) (5 points) Derive an algorithm to compute $\hat{\beta}(\lambda_1, \lambda_2)$ using nonlinear Gauss-Seidel iteration (coordinate descent).

(b) (5 points) Implement your algorithm.

```
elastic = function(y, x, lambda1, lambda2) { ... }
```

3. (40 points) Download the data file (`baseball.csv`) and its description (`baseball.txt`) from the following website.

`https://github.com/pzengauburn/STAT-7650-CompStat`

The data contain salaries for 337 baseball players in 1992, along with 27 performance statistics from 1991. The goal is to select the best subset of predictors to fit a linear regression model that minimizes AIC. The response variable is the log of salary, and the intercept term is included in all models.

(a) (5 points) Implement a local search using the steepest descent method. Start with an initial subset containing a randomly selected predictor, and perform 14 additional steps. At each step, consider the 1-neighborhood, that is, either add or remove a single predictor. The next candidate solution is chosen as the one in the neighborhood with the smallest AIC. Repeat the entire procedure 5 times. Report the best subset selected in each run.

(b) (5 points) Modify the local search in the previous question to use 2-neighborhoods, which means that at each step, two predictors can be added or removed. Compare the results with those obtained from the 1-neighborhood runs.

(c) (5 points) Implement the simulated annealing method using a 1-neighborhood. The cooling schedule consists of 15 stages, with stage lengths of 60 for the first 5 stages, 120 for the next 5, and 220 for the final 5 stages. The proposal distribution is discrete uniform over the neighborhood. Update the temperature according to $\alpha(\tau_{j-1}) = 0.9\tau_{j-1}$. Run the algorithm with starting temperatures $\tau_0 = 1$ and $\tau_0 = 6$. Report your results.

(d) (5 points) In the implementation of simulated annealing, set $m_j = 1$ for all j and update the temperature using $\alpha(\tau_{j-1}) = \tau_{j-1}/(1 + 0.01\tau_{j-1})$. To be consistent with the previous question, run a total of 2000 steps. Report your results and compare them with those from the previous question.

(e) (5 points) Implement a genetic algorithm with 100 generations, each of size $P = 20$. The initial generation is selected completely at random. Use the rank-based fitness function. During breeding, select one parent with probability proportional to its fitness, and select the other parent completely at random. Use simple (single-point) crossover: choose a random position between two adjacent loci and exchange the chromosomes after this position. Keep only one offspring resulting from each crossover. Set the mutation rate to 0.01. Report your result.

(f) (5 points) Test your implementation with different mutation rates (0.005 and 0.02) and different generation sizes ($P = 15$ and $P = 25$). Compare the results cross these settings.

(g) (5 points) Maintain a generation size of $P = 20$ and a mutation rate of 0.01, but employ tournament selection with $P/5$ strata. Report the results.

(h) (5 points) Implement a tabu search algorithm. Monitor only the attributes indicating the presence or absence of each predictor. The tabu list should include moves that reverse the inclusion or removal of a predictor with a tenure of $\tau = 5$. Run the algorithm for 75 moves starting from a random subset. Use an aspiration criterion that allows a tabu move if it yields an objective function value higher than the current best value. Report your result.

4. (30 points) Load the iris data set using the following R code. The iris data set contains measurements (in centimeters) for 50 flowers from each of three iris species. We assume that the species labels are unknown and develop a Gaussian mixture model using only a single variable, petal width.

```
library(datasets)
data(iris)
x = iris$Petal.Width
```

Let $x_1, \ldots, x_n$ be iid random variables drawn from a mixture of K normal distributions.

$$x_i \sim \sum_{k=1}^{K} \pi_k N(\mu_k, \sigma_k^2), \quad i = 1, \ldots, n,$$

where π_k, μ_k, σ_k^2 are the proportion, mean, and variance for the kth sub-population. In the problem, $n = 150$ and $K = 3$. The goal is to estimate the parameters $\theta = \{(\pi_k, \mu_k, \sigma_k^2), k = 1, \ldots, K\}$ and to determine which species each flow belongs to.

(a) (5 points) Derive an EM algorithm to estimate θ.

(b) (5 points) Implement the algorithm you derived and obtain the estimate of θ.

```
Gmm = function(x, K, ...) { ... }
```

(c) (5 points) Compute the standard errors of $\hat{\theta}$.

(d) (5 points) Compute $P(Z = k \mid x_i)$ for each flower-species pair. Assign each flower to the species corresponding to the highest probability, and construct a misclassification matrix to assess the performance of classification.

(e) (5 points) Implement the MCEM (Monte Carlo EM) algorithm to estimate θ. Compare these results with those obtained in the previous question.

(f) (5 points) If the number of sub-populations K is unknown, it can be selected using AIC, where the model with the smallest AIC is preferred.

$$\text{AIC} = -2\ell(\hat{\theta}) + 2p,$$

where $\ell(\hat{\theta})$ is the maximized log-likelihood of the observed data, and p is the number of free parameters in the model. Apply your algorithm to $K = 2, 3, 4$ and compute the AIC for each case. Which model yields the smallest AIC?